
Embodied Control and Camaraderie Behavior: A Design Note

Design note: CQD-RFC-030

Status: draft

Topic: multi-humanoid coordination

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Abstract.

This design note template describes embodied control and the camaraderie behavior that coordinates a multi-humanoid surgical team, with a state model and a coordination cadence.

Disclaimer: This work is independent and not endorsed or sponsored by trial sponsors, FDA, CROs, sites, IRBs, regulators, or medical societies; and was generated using Artificial Intelligence.

Keywords

1 Motivation

A single robot concentrates every failure mode into one instrument; coordinating several under a camaraderie behavior spreads load and adds redundancy while keeping each robot’s code gated before it acts.

2 Control architecture

Control runs a 10 Hz sensing loop into a 1 Hz planner that broadcasts commands; every command is a gated interaction before it reaches an actuator.

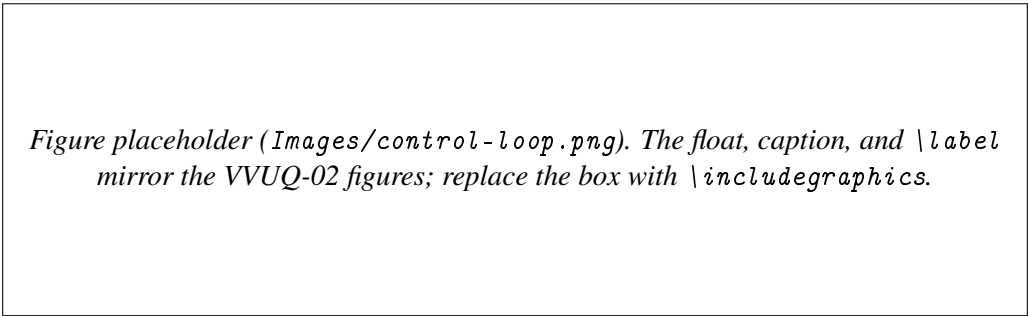


Figure 1: Figure 1. Placeholder control-loop architecture from sensing to actuation.

3 Behavior model

The behavior is a small state model.

Lead holds the active interaction and decides escalation

Assist mirrors the Lead and is ready to take over

Reserve monitors and documents, ready to promote

4 Camaraderie coordination

Roles rotate on a fixed cadence so no robot is a single point of failure, and hand-off only occurs between gated states.

Role	Cadence	Function
Lead	active	holds the gated interaction
Assist	1 Hz	ready hand-off target
Reserve	monitor	documents and promotes

Table 1: Table 1. Role, cadence, and function (modeled on VVUQ-05 Table 1).

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Cite This Note

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Data Availability

Supporting records are openly available in the [kevinkawchak/cancer-automated](#) and [kevinkawchak/physical-ai-oncology-trials](#) GitHub repositories and are archived on Zenodo. All data sets are synthetic or illustrative and reproducible from a deterministic seed; no patient-identifiable information is included.

References

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